

# Using Parallel Ports to Control Devices

Peripheral devices are connected to a computer through interfaces called ports, which could be serial or parallel. A parallel port is a parallel physical communication interface. Let's find out how to use a parallel port to control our devices. The code mentioned in this article can be used easily by anyone familiar with the Linux system and with writing a character device driver.



Parallel ports were first introduced around 1970 by Centronics. The company used a 36-pin interface at that time. Later, in the 80s, IBM started its own parallel port, which was in the form we know today. It had a 25-pin interface and the connector was called DB-25. Originally, only printers were used on this port but, eventually, scanners, zip-drives, joysticks, CD ROM drives, etc, all came to be used on this port.

In this article, we are going to use a parallel port to control our devices. Out of a total of 25 pins, we can use 12 pins for our devices. Please refer to the schematic in Figure 1.

The pins that are marked O are output pins and can be used to control devices. Just to show you how to use them, this article covers the use of two devices. One device is connected to the data pins (D3) and the other device is connected to one of the control pins (Strobe), which is active-low. I have given a relay in the schematic so that you can use any device you want. Just keep in mind that the data pins will give you 3V output and the control pins will give you 5V output.

On Linux systems, we will be using the parallel port in two ways:

1. From user space using *ioctl* system calls;

Table 1: Pin descriptions

|         |                        |     |
|---------|------------------------|-----|
| 1       | Strobe (active-low)    | I/O |
| 2       | Data0                  | O   |
| 3       | D1                     | O   |
| 4       | D2                     | O   |
| 5       | D3                     | O   |
| 6       | D4                     | O   |
| 7       | D5                     | O   |
| 8       | D6                     | O   |
| 9       | D7                     | O   |
| 10      | Acknowledgement        | I   |
| 11      | Busy (active-low)      | I   |
| 12      | Paper-out              | I   |
| 13      | Select                 | I   |
| 14      | Line Feed (active-low) | I/O |
| 15      | Error                  | I   |
| 16      | Reset                  | I/O |
| 17      | Select (active-low)    | I/O |
| 18 - 25 | Ground                 |     |

2. With a device driver, which will create device nodes in the */dev* folder; then, from the user space, we just need to write 1 or 0 to control the devices.